Grade 2
Ontario Curriculum
Spatial Sense

Mirror Mirror

Unit Overview: Mirror Mirror

Grade: Grade 2 | Strand: Spatial Sense

Target Expectations: E1.1, E1.2, E1.3, E1.4, E1.5

Learning Goal: Children will sort by sides/angles/symmetry, compose/decompose with area conservation, identify congruent shapes, create maps, and describe positions and movements.

Lesson	Lesson Title	Learning Goal
Day 1	**Sort by Sides and Symmetry**	I can sort shapes by sides, angles, and symmetry.
Day 2	**Compose, Decompose, Conserve Area**	I can show area stays the same when parts are rearranged.
Day 3	**Congruent Shapes**	I can identify congruent shapes.
Day 4	**Make a Map**	I can make a simple map of a familiar place.
Day 5	**Mirror Maze Capstone**	I can describe positions and movements on a map.

Lesson 1: Sort by Sides and Symmetry

Goal: Sort 2D shapes by SIDES + SYMMETRY: count edges, fold to test symmetry.
Expectation: E1.1

Learning Goal: "I can sort shapes by sides, angles, and symmetry."

Materials: M1_shape_set, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Symmy Counts the Shape

Symmy: This SQUARE has 4 sides, 4 angles, and 4 lines of symmetry! Let's count them all! Symmy opens: 'Count the SIDES — fold the SHAPE — match the HALVES!'

Bring children to the carpet. Symmy and Flippy puppets. 'Friends — I'm Symmy the Symmetry Squirrel and this is Flippy the Folding Frog. Today we sort shapes by THREE things: SIDES, ANGLES, and LINES OF SYMMETRY.'

2. Action (25 mins): Sort Stations

- Whole group: count sides/angles/symmetry on 4 shapes.
- Three stations.
- Stretch at C: 2-attribute sort (4-sided AND symmetric).
- Discuss: same shape, different sorting attribute = different piles.
- Circulate to record diagnostic evidence.
- STRATEGY COMPARE: Two pairs share their fold-test methods.

3. Consolidation (13 mins): Symmy's Sort Brag

Discuss:
- What's a SIDE?
- What's an ANGLE?
- What's a SYMMETRY LINE?
- How many lines does a SQUARE have?
- Symmy's preview: tomorrow Flippy REARRANGES these shapes — but watch what stays the SAME (Lesson 2 — E1.2 area conservation).

Exit: Symmy's refrain — class chants: 'Count the SIDES — fold the SHAPE — match the HALVES!' (3x).

Lesson 2: Compose, Decompose, Conserve Area

Goal: When shapes are REARRANGED, the AREA stays the SAME. Decompose, rearrange, recompose.
Expectation: E1.2

Learning Goal: "I can show area stays the same when parts are rearranged."
Materials: M2_pattern_blocks, M8_area_chart, M10_unit_poster

1. Minds On (12 mins): Flippy Rearranges Tiles

Flippy: I have 6 tiles. I make a rectangle — area = 6. Now I REARRANGE into an L-shape — STILL 6! Area stays the SAME! Flippy opens: 'Same TILES, SAME area — Flippy's rule!' Yesterday Symmy SORTED shapes — today we DECOMPOSE them.

Bring children to the carpet. Symmy and Flippy puppets. 'Friends — yesterday sides and symmetry. Today COMPOSE, DECOMPOSE, and AREA CONSERVATION.'

2. Action (25 mins): Area Stations

- Whole group: 6 tiles arranged 3 different ways.
- Three stations.
- Stretch at C: rearrange 12 tiles into 4 different shapes.
- Discuss area conservation rule.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two pairs decompose a hexagon — Pair A into triangles, Pair B into a rectangle + two triangles.

3. Consolidation (13 mins): Flippy's Area Brag

Discuss:
- What does COMPOSE mean?
- What does DECOMPOSE mean?
- What is AREA?
- Why does area stay the same when we rearrange?

Exit: Flippy's refrain — class chants: 'Same TILES, SAME area — Flippy's rule!' (3x).

Lesson 3: Congruent Shapes

Goal: CONGRUENT shapes are EXACTLY the same — same sides, same angles. We MATCH to test.
Expectation: E1.3

Learning Goal: "I can identify congruent shapes."

Materials: M3_congruent_cards, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Symmy Matches Pairs

Symmy: I have TWO triangles. Are they CONGRUENT? Let's MATCH them — sit one ON TOP. If they fit perfectly, YES! Symmy opens: 'Same SIDE, same ANGLE — CONGRUENT!'

Bring children to the carpet. Symmy and Flippy puppets. 'Friends — yesterday compose/decompose. Today CONGRUENT shapes.'

2. Action (25 mins): Congruent Stations

- Whole group: match 4 pairs.
- Three stations.
- Stretch at C: include rotated shapes.
- Discuss rotation OK, size matters.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two children test congruence three ways — (a) eyeball, (b) overlap, (c) measure sides.

3. Consolidation (13 mins): Symmy's Congruent Brag

Discuss:

- What does CONGRUENT mean?
- How do we test?
- Does rotation matter?
- Does size matter?

Exit: Symmy's congruence refrain — 'SIT one on TOP — does it MATCH? Same SIDE, same ANGLE — CONGRUENT!' Class chants 3x.

Lesson 4: Make a Map

Goal: A MAP shows places from above. We use SYMBOLS for objects.

Expectation: E1.4

Learning Goal: "I can make a simple map of a familiar place."

Materials: M4_map_template, M5_symbol_cards, M10_unit_poster

1. Minds On (12 mins): Flippy Maps the Classroom

Flippy: A MAP shows our classroom from ABOVE. The desk is a square. The door is a rectangle. The carpet is a circle! Symmy + Flippy open: 'Top-down, top-down, that's how MAPS look!'

Bring children to the carpet. Symmy and Flippy puppets. 'Friends — yesterday congruence. Today we make a MAP.'

2. Action (25 mins): Map Stations

- Whole group: map our classroom together.
- Three stations.
- Stretch at C: include compass rose (N/S/E/W).
- Discuss bird's-eye view.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two pairs share their bedroom maps and compare how they chose symbols.

3. Consolidation (13 mins): Flippy's Map Brag

Discuss:

- What is a MAP?
- Why from ABOVE?
- What are SYMBOLS?
- What's important to LABEL?

Exit: Symmy + Flippy map refrain — 'Top-down, top-down, that's how MAPS look!' Class chants 3x.

Lesson 5: Mirror Maze Capstone

Goal: We DESCRIBE positions and MOVEMENTS to navigate a maze. Forward, turn, reach the goal.
Expectation: E1.5

Learning Goal: "I can describe positions and movements on a map."

Materials: M1_shape_set, M2_pattern_blocks, M3_congruent_cards, M4_map_template

1. Minds On (12 mins): Symmy and Flippy's Maze

Symmy: Welcome to the MIRROR MAZE! Use directions to navigate. Forward 3, turn right, forward 2 — REACH THE GOAL! Symmy + Flippy open: 'Sort, AREA, CONGRUENT, MAP!'

Bring children to the carpet. Symmy and Flippy puppets in capes. 'Friends — today is our CAPSTONE. We use ALL we know: sides + symmetry, compose + area, congruent, maps, AND directions.'

2. Action (25 mins): The Mirror Maze

- Hand out templates.
- Stations 1-5: shapes, area, congruent, map, maze.
- Stretch: 'big idea' conclusion at the end.
- Discuss as students finish.
- Circulate to record summative evidence.

3. Consolidation (13 mins): Symmy's Maze Brag

Discuss:

- Which station was your favourite?
- How did you write directions?
- What did you map?
- What's CONGRUENT?

Exit: Symmy + Flippy capstone refrain — 'Sort, AREA, CONGRUENT, MAP — Symmy + Flippy show the WAY!' Class chants 3x.

Name: ______________________ Date: ______________________

Symmy's Shape Counter

Learning Goal: I can sort shapes by sides, angles, and symmetry.

Mission

1. Look at each shape.
2. Draw lines of symmetry on each shape.
3. Sort 6 shapes by NUMBER OF SIDES.

Part 3: Sort by sides.

Sort 6 shapes by NUMBER OF SIDES.

(3 parts on the printable worksheet)

Name: ______________________________ Date: ___________________________

Flippy's Area Lab

Learning Goal: I can show area stays the same when parts are rearranged.

Instructions for the Student:

1. Count the tiles in each arrangement.
2. A hexagon decomposes into 6 triangles.
3. Rearrange 8 tiles into 3 different shapes.

Part 2: Decompose hexagon.

A hexagon decomposes into 6 triangles. Show area = 6.

(3 parts on the printable worksheet)

Name: ______________________ Date: ______________________

Symmy's Congruent Match

Learning Goal: I can identify congruent shapes.

Mission

1. Look at 5 pairs.
2. For each shape, find the CONGRUENT partner from a row of 5.
3. Draw a shape CONGRUENT to the given one.

Part 3: Draw congruent.

Draw a shape CONGRUENT to the given one.

(3 parts on the printable worksheet)

Name: ______________________________ Date: ______________________________

Flippy's Map Maker

Learning Goal: I can make a simple map of a familiar place.

Instructions for the Student:

1. Look at the map.
2. Draw a top-down map of your bedroom.
3. Pick a place at home (kitchen, yard).

Part 2: Map your bedroom.

Draw a top-down map of your bedroom. Use symbols.

(3 parts on the printable worksheet)

Name: ______________________ Date: ______________________

Symmy's Mirror Maze

Learning Goal: I can describe positions and movements on a map.

Instructions for the Student:

1. Count sides + lines of symmetry for 3 shapes.
2. Match 3 pairs as congruent or NOT.
3. Look at the maze.

Part 3: Maze directions.

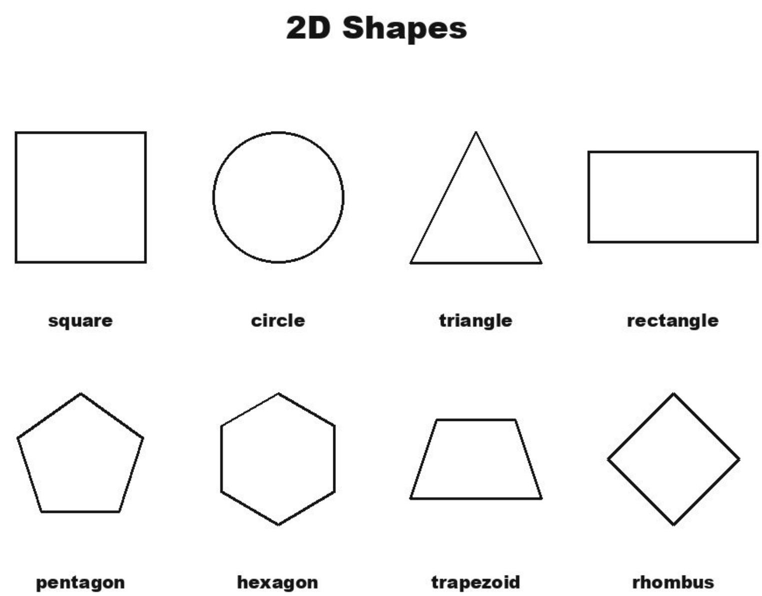
Look at the maze. Write 3 directions to navigate from START to GOAL.

(5 parts on the printable worksheet)

Sortable 2D Shape Set

(M1_shape_set)

30 paper 2D shapes for sorting by sides/angles/symmetry.



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 2
- Laminate: yes

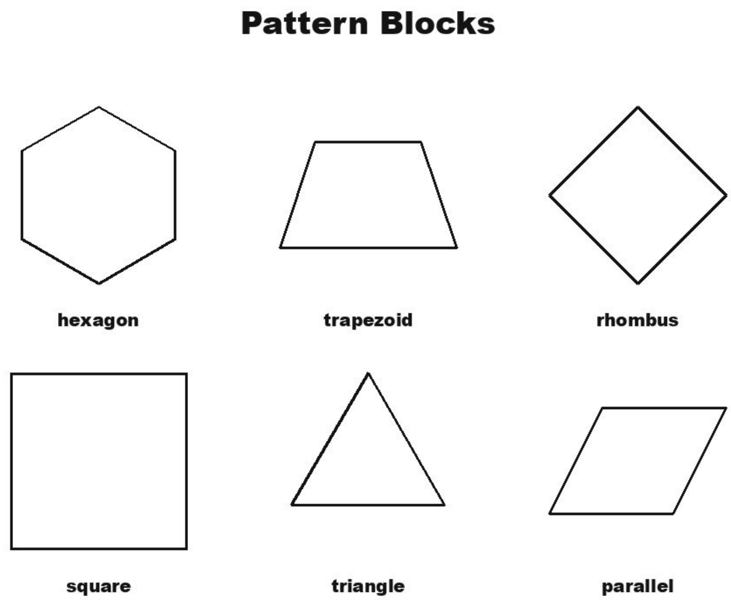
USE IN CLASS
- Quantity: 10 sets
- Lessons: 1, 5
- Prep: ~25 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 30 shapes per set.
4. Store in labelled bag.

Pattern Block Set

(M2_pattern_blocks)

60 paper pattern blocks for compose/decompose.



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 2
- Laminate: yes

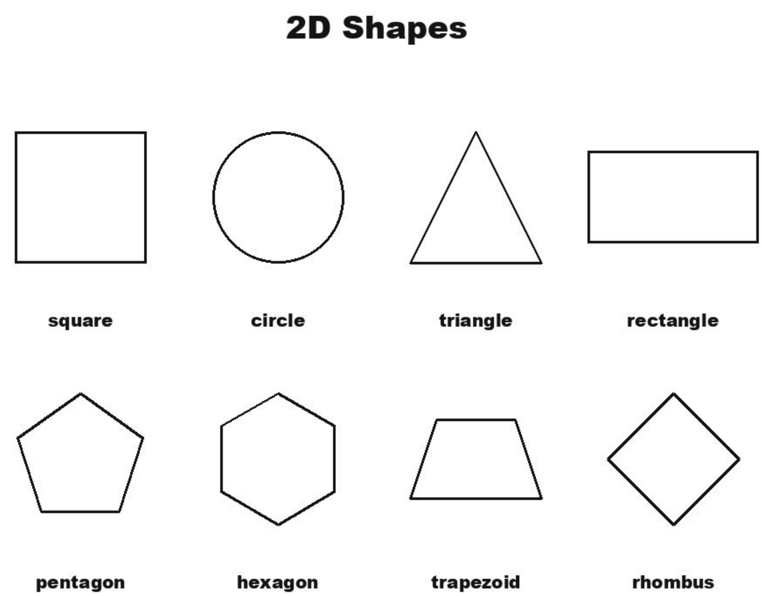
USE IN CLASS
- Quantity: 10 sets
- Lessons: 2, 5
- Prep: ~30 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 60 blocks per set.
4. Store in labelled bag.

Congruent Pair Cards

(M3_congruent_cards)

16 paper shape cards in 8 congruent pairs.



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

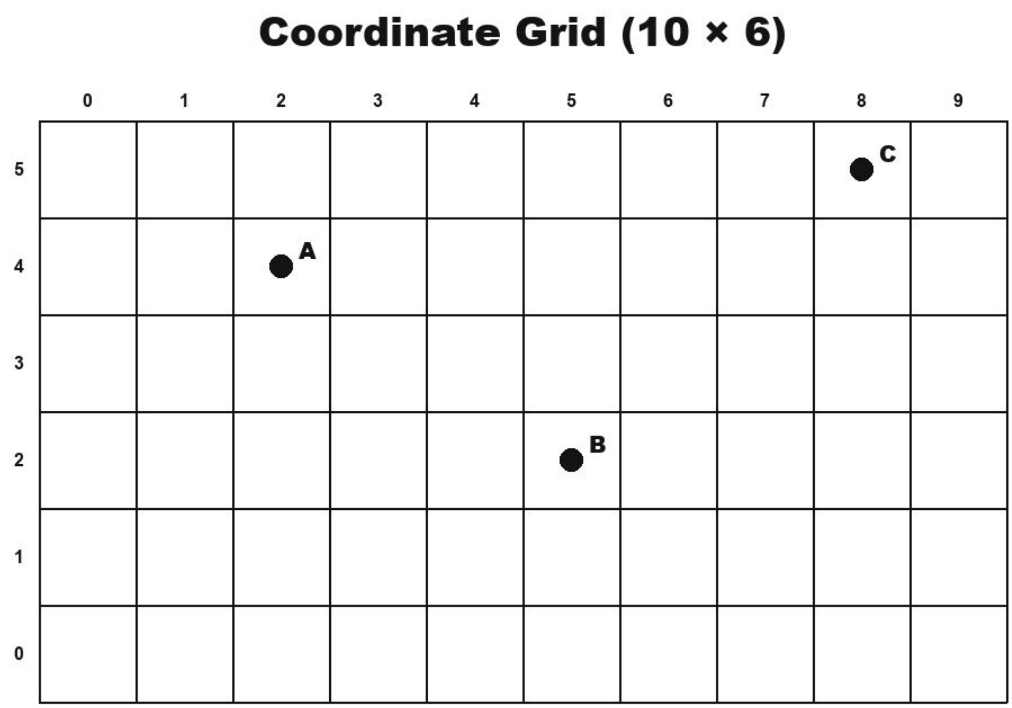
USE IN CLASS
- Quantity: 10 sets
- Lessons: 3, 5
- Prep: ~15 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 16 cards per set.
4. Store in labelled bag.

Map Template

(M4_map_template)

Letter-size map grid for drawing.



PRINT SPECS
Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

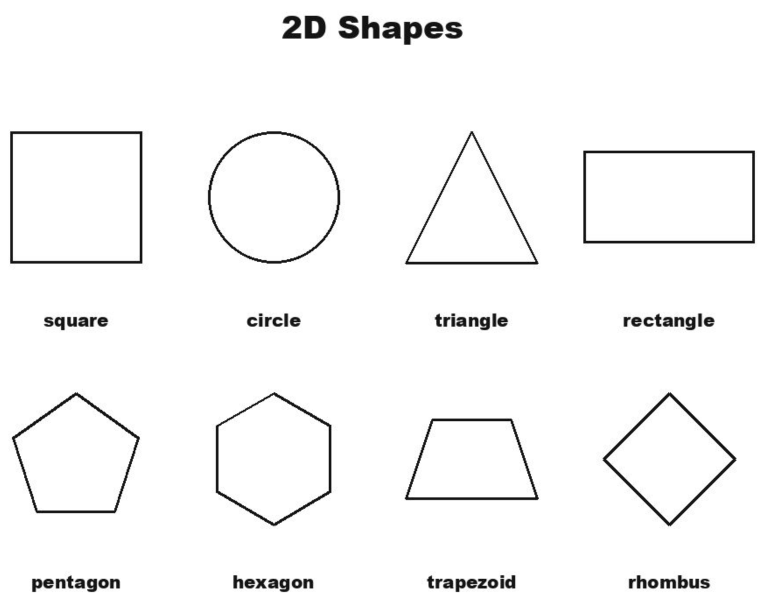
USE IN CLASS
Quantity: 10-12 templates
Lessons: 4, 5
Prep: ~10 min

TEACHER PREP
1. Print 12 templates.
2. Laminate for dry-erase.
3. Distribute one per pair.

Map Symbol Cards

(M5_symbol_cards)

12 map-symbol cards for objects (bed, desk, tree, etc.).



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

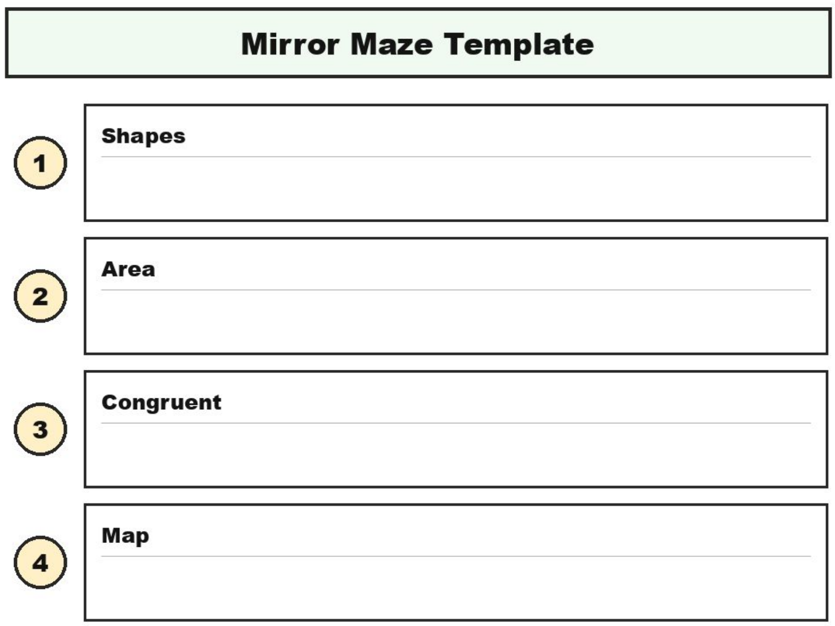
USE IN CLASS
Quantity: 10 sets
Lessons: 4, 5
Prep: ~15 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

Mirror Maze Template

(M6_capstone_template)

Letter-size capstone with maze, position lines, direction lines.



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: no

USE IN CLASS
Quantity: 25 templates
Lessons: 5
Prep: ~5 min

TEACHER PREP
1. Print 25 templates.
2. Stack near teacher table.
3. No lamination.

Shape Attribute Chart

(M7_attribute_chart)

Anchor chart of sides/angles/symmetry for common shapes.

Shape Attributes

- **Sides — straight or curved?**
- **Corners — how many?**
- **Equal sides?**
- **Right angles?**
- **Symmetry — does it match?**

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

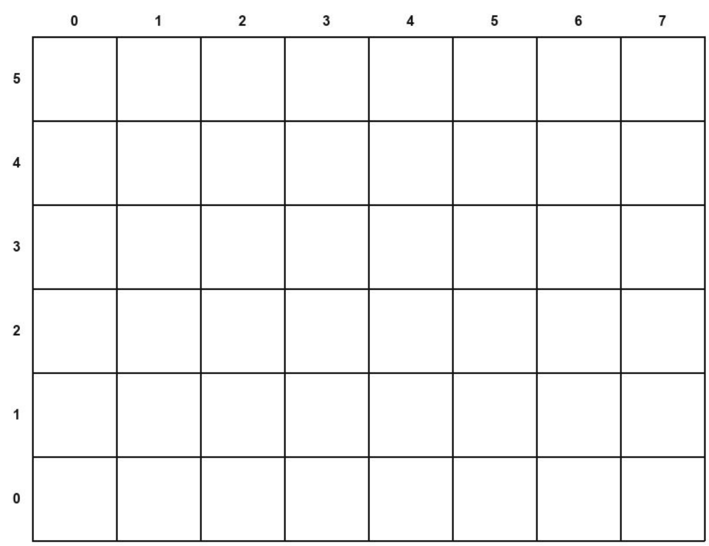
USE IN CLASS
Quantity: 1 chart
Lessons: 1, 3, 5
Prep: ~10 min

TEACHER PREP
1. Print 1 copy.
2. Laminate.
3. Tape to front wall before Lesson 1.

Area Conservation Chart

(M8_area_chart)

Anchor chart showing area stays the same when rearranged.



Coordinate Grid (8 × 6)

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

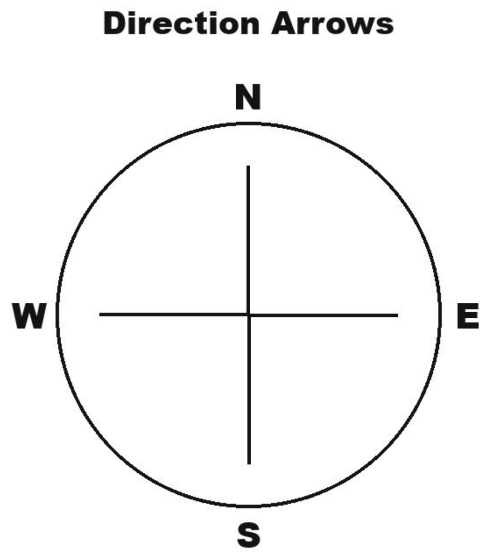
USE IN CLASS
- Quantity: 1 chart
- Lessons: 2, 5
- Prep: ~10 min

TEACHER PREP
1. Print 1 copy.
2. Laminate.
3. Tape to front wall.

Direction Cards

(M9_direction_cards)

12 cards (forward/back/left/right with numbers).



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

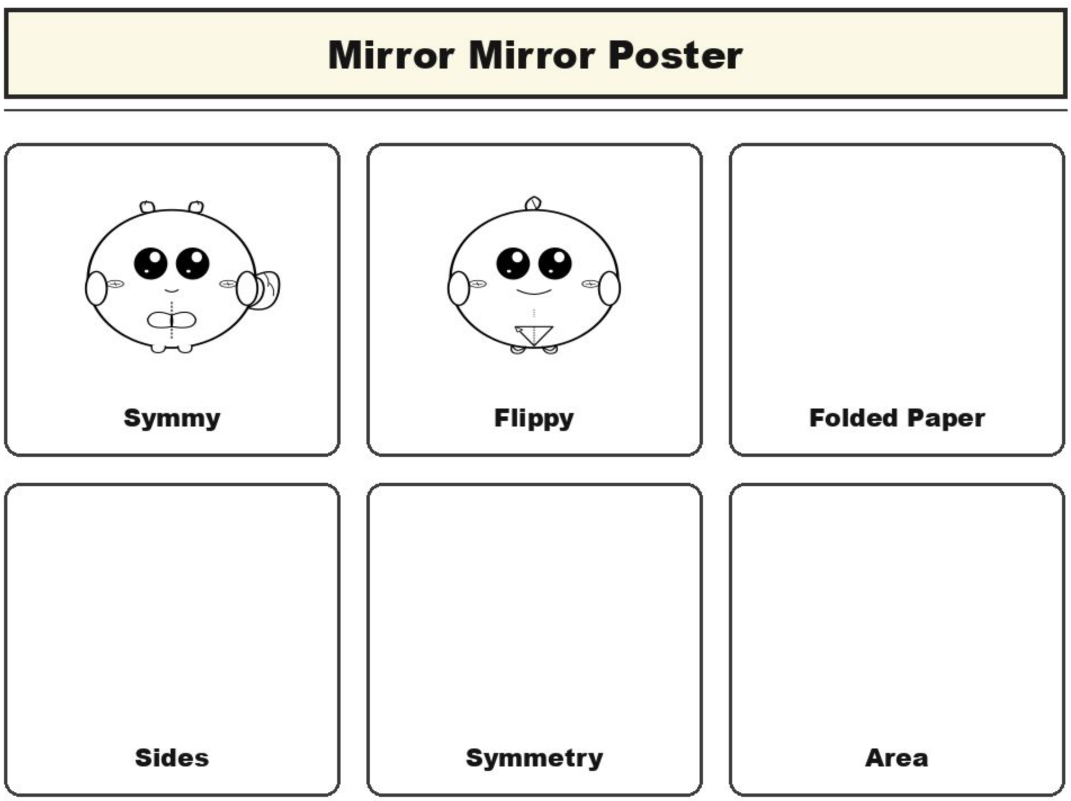
USE IN CLASS
- Quantity: 10 sets
- Lessons: 5
- Prep: ~15 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

Mirror Mirror Poster

(M10_unit_poster)

Anchor poster of Symmy and Flippy.



PRINT SPECS
- Size: letter landscape
- Colour: bw
- Pages per set: 1
- Laminate: yes

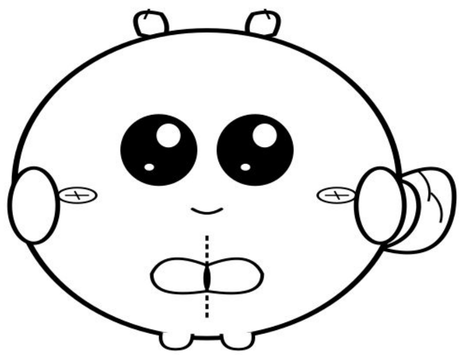
USE IN CLASS
- Quantity: 1 poster
- Lessons: 1, 2, 3, 4, 5
- Prep: ~10 min

TEACHER PREP
1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

Symmy Symmetry Squirrel Puppet

(char_symmy_puppet)

Paper puppet for teacher use in all 5 lessons.



Symmy the Symmetry Squirrel

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

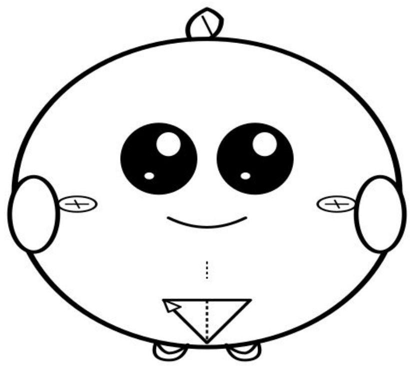
USE IN CLASS
- Quantity: 1 puppet
- Lessons: 1, 2, 3, 4, 5
- Prep: ~12 min

TEACHER PREP
1. Print 1 copy.
2. Cut around silhouette + tab.
3. Laminate.
4. Glue popsicle stick.

Flippy the Folding Frog Puppet

(char_flippy_puppet)

Paper puppet for teacher use in Lessons 2-5.



Flippy the Folding Frog

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS
- Quantity: 1 puppet
- Lessons: 2, 3, 4, 5
- Prep: ~12 min

TEACHER PREP
1. Print 1 copy.
2. Cut around silhouette + tab.
3. Laminate.
4. Glue popsicle stick.

Name: ______________________ Date: ______________________

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
E1.1 sort and identify two-dimensional shapes by comparing number of sides, side lengths, angles, and number of lines of symmetry	Conflates sides and angles. Cannot identify symmetry lines.	Counts sides correctly with prompts. Finds 1-2 symmetry lines on simple shapes.	Independently counts sides, angles, and finds all symmetry lines on common shapes.	Notices regular shapes have side count = angle count = symmetry count and articulates this.
E1.2 compose and decompose two-dimensional shapes, and show that the area of a shape remains constant regardless of how its parts are	Believes spread shapes have more area. Cannot decompose.	Counts tiles to compare area. Decomposes simple shapes.	Independently decomposes shapes and shows area unchanged regardless of arrangement.	Predicts area before counting. Articulates 'same tiles = same area'.
E1.3 identify congruent lengths and angles in two-dimensional shapes by mentally and physically matching them, and determine if the	Cannot test congruence. Conflates similar with congruent.	Tests congruence by matching with prompts.	Independently tests congruence including rotated shapes.	Articulates that congruent means exactly same — sides and angles preserved under rotation.
E1.4 create and interpret simple maps of familiar places	Draws side view instead of top-down.	Draws top-down with prompts. Labels some objects.	Independently draws top-down map with symbols and labels.	Adds compass rose, scale, or legend to the map.
E1.5 describe the relative positions of several objects and the movements needed to get from one object to	Vague directions ('go that way'). No numbers.	Uses verb without numbers.	Independently writes verb + number directions to navigate.	Plans multi-step routes and describes positions in relation to multiple landmarks.

Junior Spatial Detective Certificate (G2)

This certificate is proudly presented to:

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For successfully completing the 5-Day Mirror Mirror unit and demonstrating the skills of a G2 Junior Spatial Detective. Symmy the Symmetry Squirrel congratulates the student.

By completing this unit, this student has demonstrated:

- Counting sides, angles, and symmetry lines
- Decomposing shapes and conserving area
- Identifying congruent shapes
- Making maps and writing directions

Marketplace Listing

A 5-day Grade 2 unit on spatial sense framed around Symmy the Symmetry Squirrel and Flippy the Folding Frog.

Price: $8.99 CAD
Pages: ~45
Classroom time: 250 min
Teacher prep: ~200 min

Top tags: Grade 2, math, spatial, geometry, symmetry, congruent, area, maps

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan with 12 print-ready assets (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)